

Homework 1 CS5033 Machine Learning Fundamentals

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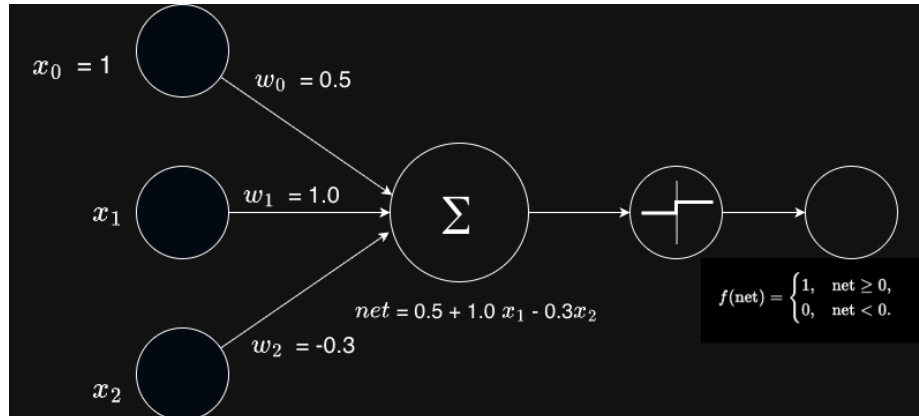
Due Fri Sept 18

1 Part 1- AN Representation

1. Consider a single AN used for classification in a 2D space with an augmented vector as discussed in class. That is, this AN has a constant input of 1 as its zeroth input (which is multiplied by w_0) as well as inputs x_1 and x_2 , which are multiplied by w_1 and w_2 , respectively. This AN is a summation unit (SU), and its activation function f is a step function.

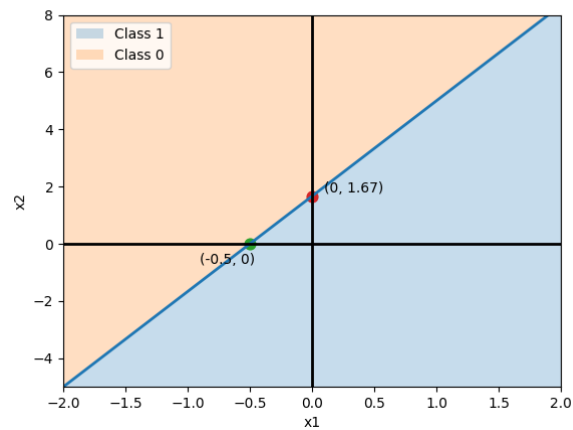
$$f(\text{net}) = \begin{cases} 1, & \text{net} \geq 0, \\ 0, & \text{net} < 0. \end{cases}$$

Given the weights $w_0 = 0.5$, $w_1 = 1.0$, $w_2 = -0.3$. draw this AN.

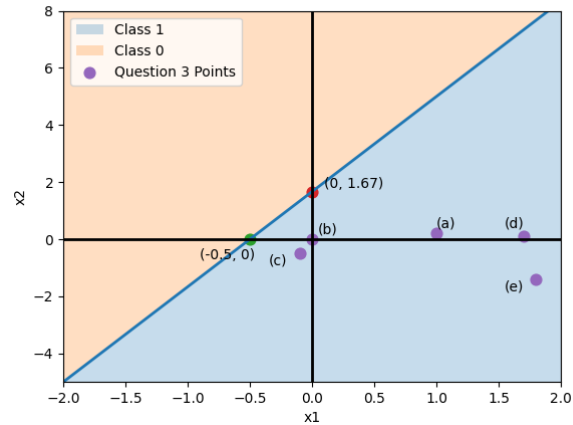


2. Draw (on graph or engineering paper or by using software) the decision boundary encoded by this AN. Be sure to indicate the “front” (Class 1) side of the boundary.

$$\begin{aligned}
 0 &= 0.5 + x_1 - 0.3x_2 \\
 0.3x_2 &= 0.5 + x_1 \\
 x_2 &= \frac{0.5 + x_1}{0.3} \\
 (0, x_2) &= (0, 1.67) & (x_1, 0) &= (-0.5, 0)
 \end{aligned}$$



3. Add the following points on the graph you just drew, and label the class of each according to the AN:



$$\begin{aligned}
 (a) \ (1.0, 0.2) : \quad net &= 0.5 + 1.0 - 0.3(0.2) \\
 &= 1.44 \\
 &\Rightarrow \text{Class 1}
 \end{aligned}$$

$$\begin{aligned}
 (b) \ (0, 0) : \quad net &= 0.5 + 0 - 0 \\
 &= 0.50 \\
 &\Rightarrow \text{Class 1}
 \end{aligned}$$

$$\begin{aligned}
 (c) \ (-0.1, -0.5) : \quad net &= 0.5 - 0.1 - 0.3(-0.5) \\
 &= 0.55 \\
 &\Rightarrow \text{Class 1}
 \end{aligned}$$

$$\begin{aligned}
 (d) \ (1.7, 0.1) : \quad net &= 0.5 + 1.7 - 0.3(0.1) \\
 &= 2.17 \\
 &\Rightarrow \text{Class 1}
 \end{aligned}$$

$$\begin{aligned}
 (e) \ (1.8, -1.4) : \quad net &= 0.5 + 1.8 - 0.3(-1.4) \\
 &= 2.72 \\
 &\Rightarrow \text{Class 1}
 \end{aligned}$$

4. Assume that the AN's classification of each of the points above is correct. List a new labeled data item which, if added to the data set above, would necessarily cause the AN to misclassify at least one data item. That is, given this new data item, there would be no set of possible weights that would allow this AN to correctly classify all data items. Explain why the AN would necessarily misclassify at least one data item.

The point $(0, 0)$ is already in the original data set. For the current AN,

$$\begin{aligned}
 net &= 0.5 + x_1 - 0.3x_2 \\
 &= 0.5 + 0 - 0.3(0) \\
 &= 0.5
 \end{aligned}$$

Since $net \geq 0$, the point $(0, 0)$ is classified as Class 1.

If we add the same point $(0, 0)$ again but label it as Class 0, the AN would have to give two different class labels to the exact same input. This is impossible for any set of weights. Therefore, at least one of the data items must be misclassified.

5. Explain how the decision boundary for this AN would change if f were changed to be

$$f(net) = \begin{cases} 1, & net \geq 0, \\ -1, & net < 0. \end{cases}$$

instead.

Note for explanatory questions, like this one: To explain the answer, describe what the answer is—in this case, what changes (if any) there are to the decision boundary—and also why that is the correct answer. In this case that answer would be based on how the value of $f(\text{net})$ affects the decision boundary.

The decision boundary does not change because net is calculated by the weighted sum

$$\text{net} = 0.5 + \mathbf{x}_1 - 0.3\mathbf{x}_2.$$

The decision boundary is still where

$$\text{net} = 0.$$

The only thing that changes is the output, also known as the class label. Instead of giving an output of 0, it now gives an output of -1 .

6. Explain how the decision boundary for this AN would change if f were changed to be the sigmoidal logistic function

$$f(\text{net}) = \frac{1}{1 + e^{-\text{net}}}$$

instead.

It doesn't change the decision boundary; it changes the neuron output values, but it does not move the boundary. Because when $f(0)$ is calculated using the sigmoidal function, the output is 0.5. Therefore, the boundary is still at $\text{net} = 0$.

7. Explain how the classification of each of the points listed above would change if f were a sigmoidal logistic function.

$$\begin{aligned} f(\text{net}) &= \frac{1}{1 + e^{-1.44}} \\ &\approx 0.808 \\ &\Rightarrow \text{Class 1} \end{aligned}$$

$$\begin{aligned} f(\text{net}) &= \frac{1}{1 + e^{-0.50}} \\ &\approx 0.622 \\ &\Rightarrow \text{Class 1} \end{aligned}$$

$$\begin{aligned} f(\text{net}) &= \frac{1}{1 + e^{-0.55}} \\ &\approx 0.634 \\ &\Rightarrow \text{Class 1} \end{aligned}$$

$$\begin{aligned}
f(net) &= \frac{1}{1 + e^{-2.17}} \\
&\approx 0.898 \\
&\Rightarrow \text{Class 1}
\end{aligned}$$

$$\begin{aligned}
f(net) &= \frac{1}{1 + e^{-2.72}} \\
&\approx 0.938 \\
&\Rightarrow \text{Class 1}
\end{aligned}$$

Since all points are class 1 and is between 0 and 1 and all outputs are greater than 0.5 the classification didn't change

2 Part 1- AN learning

1. Consider the same AN given above in 1.1 together with the following learning rule (and ignore the points in 1.3 and 1.4 and possible modifications to the AN listed in 1.5 – 1.7):

$$\mathbf{w}_{new} = \mathbf{w}_{old} + (t - a)\mathbf{p},$$

where t is the target value for the AN, a is the actual output value of the AN (after application of the step function $f(net)$), and $\mathbf{p}$ is the point provided as input to the AN. Explain how the AN weights would be updated given each of the following labeled data points, presented in the following order:

- (a) Class 0: $(-0.1, -1.0)$
- (b) Class 0: $(0.2, -0.9)$
- (c) Class 1: $(0.6, 0.8)$
- (d) Class 1: $(0.0, 0.0)$

$$\mathbf{w} = (w_0, w_1, w_2)$$

$$\mathbf{w}_{old} = (0.5, 1.0, -0.3)$$

$$\mathbf{p} = (1, x_1, x_2)$$

$$\mathbf{w}_{new}$$

$$f(net) = \begin{cases} 1, & net \geq 0 \\ 0, & net < 0 \end{cases}$$

(a) Class 0: $(-0.1, -1.0)$

$$t = 0$$

$$\mathbf{p} = (1, -0.1, -1.0)$$

$$\mathbf{w}_{old} = (0.5, 1.0, -0.3)$$

$$\begin{aligned} net &= 0.5 + (1.0)(-0.1) + (-0.3)(-1.0) \\ &= 0.7 \end{aligned}$$

$$a = 1$$

$$t - a = 0 - 1 = -1$$

$$\begin{aligned} \mathbf{w}_{new} &= (0.5, 1.0, -0.3) + (-1)(1, -0.1, -1.0) \\ &= (0.5, 1.0, -0.3) + (-1, 0.1, 1.0) \\ &= \boxed{(-0.5, 1.1, 0.7)} \end{aligned}$$

(b) Class 0: $(0.2, -0.9)$

$$t = 0$$

$$\mathbf{p} = (1, 0.2, -0.9)$$

$$\mathbf{w}_{old} = (-0.5, 1.1, 0.7)$$

$$\begin{aligned} net &= -0.5 + (1.1)(0.2) + (0.7)(-0.9) \\ &= -0.91 \end{aligned}$$

$$a = 0$$

$$t - a = 0 - 0 = 0$$

The weights stay the same.

$$\boxed{\mathbf{w}_{new} = (-0.5, 1.1, 0.7)}$$

(c) Class 1: $(0.6, 0.8)$

$$t = 1$$

$$\mathbf{p} = (1, 0.6, 0.8)$$

$$\mathbf{w}_{old} = (-0.5, 1.1, 0.7)$$

$$\begin{aligned} net &= -0.5 + (1.1)(0.6) + (0.7)(0.8) \\ &= 0.72 \end{aligned}$$

$$a = 1$$

$$t - a = 1 - 1 = 0$$

The weights stay the same.

$$\boxed{\mathbf{w}_{new} = (-0.5, 1.1, 0.7)}$$

(d) Class 1: (0.0, 0.0)

$$t = 1$$

$$\mathbf{p} = (1, 0.0, 0.0)$$

$$\mathbf{w}_{old} = (-0.5, 1.1, 0.7)$$

$$\begin{aligned} net &= -0.5 + (1.1)(0.0) + (0.7)(0.0) \\ &= -0.5 \end{aligned}$$

$$a = 0$$

$$t - a = 1 - 0 = 1$$

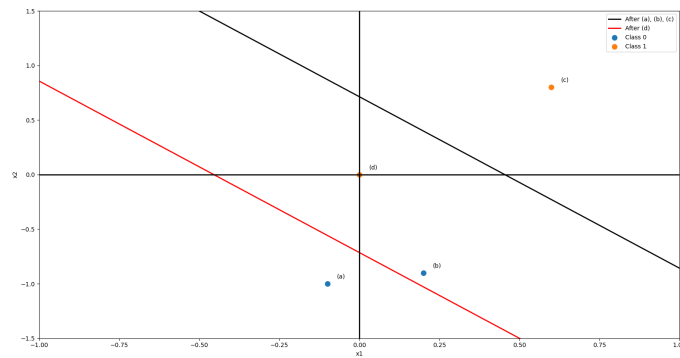
$$\begin{aligned} \mathbf{w}_{new} &= (-0.5, 1.1, 0.7) + (1)(1, 0.0, 0.0) \\ &= (-0.5, 1.1, 0.7) + (1, 0, 0) \\ &= \boxed{(0.5, 1.1, 0.7)} \end{aligned}$$

The AN only updates its weights when a point is misclassified; points a and d change the weights, whereas points b and c leave them unchanged, giving final weights $\mathbf{w}=(0.5,1.1,0.7)$

2. Draw the graph of the data and the decision boundary after each weight update for the data given in 2.1.

$$y = \frac{0.5 - 1.1x}{0.7}$$

$$y = \frac{-0.5 - 1.1x}{0.7}$$



3. Explain which of the points in 2.1 are correctly classified by this AN after all of the weight updates from one learning pass through this data, and explain what this tells us about the use of this learning rule.

$$\mathbf{w} = (0.5, 1.1, 0.7)$$

Class 0: $(-0.1, -1.0)$

$$\begin{aligned} net &= 0.5 + (1.1)(-0.1) + (0.7)(-1.0) \\ &= -0.31 \end{aligned}$$

$$-0.31 < 0 \Rightarrow \text{Class 0}$$

Correct

Class 0: $(0.2, -0.9)$

$$\begin{aligned} net &= 0.5 + (1.1)(0.2) + (0.7)(-0.9) \\ &= 0.09 \end{aligned}$$

$$0.09 \geq 0 \Rightarrow \text{Class 1}$$

Incorrect

Class 1: $(0.6, 0.8)$

$$\begin{aligned} net &= 0.5 + (1.1)(0.6) + (0.7)(0.8) \\ &= 1.72 \end{aligned}$$

$$1.72 \geq 0 \Rightarrow \text{Class 1}$$

Correct

Class 1: (0.0,0.0)

$$\begin{aligned} net &= 0.5 + (1.1)(0.0) + (0.7)(0.0) \\ &= 0.5 \end{aligned}$$

$$0.5 \geq 0 \Rightarrow \text{Class 1}$$

Correct

a, c, and d were correctly classified, but b was misclassified. This point to the fact that a point can be classified correctly at one step and then become incorrect after the weights are updated later. Because of this, the AN may need multiple passes through the data before all points are classified correctly.

4. Explain how many more passes through the data are needed before all of the points from 2.1 are correctly classified by this AN. You may answer this question on a scale of “one, two, many.” That is, if all of the points are not correctly classified after two more passes through the data, you may stop at a total of three passes and report that it still has not correctly classified all of the points after three passes. Show your work.

We know that point (a) is correctly classified, but point (b) is not, so we need to update the weights using

$$\mathbf{w}_{new} = \mathbf{w}_{old} + (t - a)\mathbf{p}$$

For point (b),

$$\begin{aligned} \mathbf{w}_{new} &= (0.5, 1.1, 0.7) + (-1)(1, 0.2, -0.9) \\ &= \boxed{(-0.5, 0.9, 1.6)} \end{aligned}$$

For point (c),

$$\begin{aligned} net &= -0.5 + 0.9(0.6) + 1.6(0.8) \\ &= 1.32 \end{aligned}$$

Point (c) is correctly classified.

For point (d),

$$\begin{aligned} net &= -0.5 + 0.9(0) + 1.6(0) \\ &= -0.5 \end{aligned}$$

Point (d) is incorrectly classified, so we update the weights again:

$$\begin{aligned} \mathbf{w}_{new} &= (-0.5, 0.9, 1.6) + (1)(1, 0, 0) \\ &= \boxed{(0.5, 0.9, 1.6)} \\ \mathbf{w} &= (0.5, 0.9, 1.6) \end{aligned}$$

$$\begin{aligned} (a) \quad net &= 0.5 + 0.9(-0.1) + 1.6(-1.0) \\ &= -1.19 \end{aligned}$$

Class 0, Correct

$$\begin{aligned} (b) \quad net &= 0.5 + 0.9(0.2) + 1.6(-0.9) \\ &= -0.76 \end{aligned}$$

Class 0, Correct

$$\begin{aligned} (c) \quad net &= 0.5 + 0.9(0.6) + 1.6(0.8) \\ &= 2.32 \end{aligned}$$

Class 1, Correct

$$\begin{aligned} (d) \quad net &= 0.5 + 0.9(0) + 1.6(0) \\ &= 0.5 \end{aligned}$$

Class 1, Correct

All four points are correctly after one more pass through the data.

5. Explain the likely effect of introducing a learning-rate parameter (η) into the equation given above.

Adding a learning rate η controls how much the weights change during each update. The new rule becomes

$$\mathbf{w}_{new} = \mathbf{w}_{old} + \eta(t - a)\mathbf{p}.$$

If η is small, the weights change by a smaller amount, so learning may take longer. If η is large, the weights change by a larger amount, which may make learning faster but can also make the updates too large.

Therefore, the learning rate controls the size of each step the AN takes when updating its weights.